

Date:

Roll No.

ODD SEMESTER B.Tech (ECE, EVDT)

MID-SEMESTER EXAMINATION, SEPTEMBER 2025

Course Code- ECECC302/ VTECC302

E1FCE302

Time: 1:30 Hrs

Course Title- Signals & Systems

Max Marks: 20

Attempt all questions. Missing data/information (If any) may be suitably assumed & mentioned in the answer.

Q.No.	Questions	Marks	CO
Q 1	Given that $x(t)$ and $y(t)$ are periodic with time period equal to T ? Are the following signals periodic? If yes what is the time period. i. $ax(t)+by(t)$, here a & b are constants. ii. $x(t).y(2t)$ iii. $x(mt)+y(nt)$, here m & n are integers.	03	CO1
	b Compute the integral $\int_{-\infty}^{\infty} \{e^{-t} \cdot \sum_{n=0}^{\infty} \delta(t-n)\} dt$	01	CO1
Q 2	Consider a discrete-time system whose input $x[n]$ and output $y[n]$ are related as $y[n] = \sum_{k=-\infty}^{\infty} 2^{k-n} x[k+1]$. Is the system LTI? Is the system causal, stable?	03	CO1
	b Determine whether $x[n] = \cos\left(\frac{\pi}{8} n^2\right)$ is an Energy or Power Signal.	01	CO1
Q 3	Compute and plot the convolution of $x[n]$ and $h[n]$. Where, the input $x[n] = \left(\frac{1}{3}\right)^{-n} u[-n]$ and impulse response $h[n] = u[2-n]$. Compute the system output when the input is $x_2[n] = \left(\frac{1}{3}\right)^{1-n} u[1-n]$.	03	CO2
	b $y[n] = x[n+1] - 2x[n] + x[n-1]$. Compute the impulse response $h[n]$, and hence comment on the stability of the system.	01	CO1
Q 4	Let $h(t)$ be the impulse response of a causal and stable continuous time LTI system. Comment on the stability and causality of the cascade of two LTI systems with impulse responses $h(t)$ and $h(-t)$.	03	CO2
	b The unit step response of an LTI system is $s(t) = (1 - e^{-2t})u(t)$. What is the response to an input $x(t) = 4u(t+2) - u(t+1)$?	01	CO2
Q 5	Determine the Fourier series coefficients of a periodic signal $x(t)$ which has a time period 4. $x(t) = \begin{cases} -1, & -1 \leq t < 0 \\ 2, & 0 \leq t < 2 \\ -1, & 2 \leq t < 3 \end{cases}$	03	CO2
	b Determine the DTFS coefficients of $x(n) = 1 + \cos(n\pi/12 + 3\pi/8)$	01	CO2